

Infinitesimal Calculus 2, Moed A

2024, Reichman university.

Do not write at the back of the papers!!

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Instructions:

1. Duration: 3 hours.

Instructions:

The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

Behatzlacha!!!

1. (25 points)

- 1.1. (15 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose that the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is a Leibniz series. Prove that it converges to a value $s \in \mathbb{R}$, and that $a_1 - a_2 \leq s \leq a_1$.

Solution:

See lectures.

- 1.2. (10 points) Compute the following definite integral:

$$\int_0^1 \arccos(x) dx$$

Solution: We use integration by parts,

$$\begin{aligned} \int_0^1 \arccos(x) dx &\stackrel{u'=1, v=\arccos(x)}{=} [x \cdot \arccos(x)]_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}} dx \stackrel{t=1-x^2 \Rightarrow dt=-2xdx}{=} 0 - \frac{1}{2} \int_1^0 \frac{dt}{\sqrt{t}} \\ &= \frac{1}{2} \int_0^1 \frac{dt}{\sqrt{t}} \stackrel{\text{formula}}{=} \frac{1}{2} \cdot \frac{1}{1-\frac{1}{2}} = 1 \end{aligned}$$

We now give an alternative solution: we use the method of substitution: As $\sin(t) \geq 0$ for every $t \in [0, \frac{\pi}{2}]$ and as $\cos^2(t) + \sin^2(t) = 1$ then $\sin(t) = \sqrt{1 - \cos^2(t)}$ for every $t \in [0, \frac{\pi}{2}]$. We now compute

$$\begin{aligned} \int_0^1 \arccos(x) dx &\stackrel{t=\arccos(x) \rightarrow dt = -\frac{dx}{\sqrt{1-x^2}}}{=} - \int_{\frac{\pi}{2}}^0 t \sqrt{1-x^2} dt = \int_0^{\frac{\pi}{2}} t \sqrt{1-\cos^2(t)} dt \\ &= \int_0^{\frac{\pi}{2}} t \sin(t) dt \stackrel{u'=\sin(t), v=t}{=} [-t \cos(t)]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos(t) dt = \int_0^{\frac{\pi}{2}} \cos(t) dt = [\sin(t)]_0^{\frac{\pi}{2}} = 1 \end{aligned}$$

2. (25 points)

- 2.1. (18 points)

- 2.1.1. (5 points) Prove that the domain of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ is the interval $(-1, 1]$.

Solution: We compute the convergence radius using the root formula,

$$R = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{|a_n|}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{\frac{1}{n}}} = \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

Thus the series converges for every $-1 < x < 1$ and diverges for every $x > 1$ and $x < -1$. We now check the endpoints: the series is convergent for $x = 1$ (Leibniz) and the series is divergent for $x = -1$ (minus Harmonic),

We conclude that the domain of convergence of the power series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$ is the interval $(-1, 1]$.

- 2.1.2. (13 points) Prove that for every $-1 < x \leq 1$ we have

$$\ln |1+x| = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

Solution: See lectures.

- 2.2. (7 points) Prove or disprove the following statement: the series $1 - 1 + 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} + \frac{1}{3} - \frac{1}{3} + \frac{1}{3} + \dots$ is convergent.

Solution:

Solution: False. ATC that the series is convergent. Applying parentheses on every 3 terms, then the resulting series is convergent by the parentheses test. But

$$(1 - 1 + 1) + \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{3} + \frac{1}{3}\right) + \cdots = 1 + \frac{1}{2} + \frac{1}{3} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Contradiction.

3. (25 points)

3.1. (17 points) Let f be a function that is defined on $[0, 1]$.

3.1.1. (2 points) Define the term: f is differentiable on $[0, 1]$. Note: you do not have to use the $\epsilon - \delta$ language.

Solution: We say that f is differentiable on $[0, 1]$ if for every $x_0 \in (0, 1)$, the limit $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exists, and also the right-sided limit $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x}$ exists.

3.1.2. (15 points) Suppose that f is continuous on $[0, 1]$ and differentiable on $(0, 1)$. Suppose also that $f(0) = 0$ and that $f'(0) = f(1)$. Prove that there exists a $0 < c < 1$ for which $f'(c) = \frac{f(c)}{c}$.

Solution: Consider the function that is defined by $g(x) = \begin{cases} \frac{f(x)}{x} & 0 < x \leq 1 \\ f(1) & x = 0 \end{cases}$ for every $0 \leq x \leq 1$. Note that g is

continuous on $(0, 1]$ by AOC. Note also that $\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} \frac{f(x)}{x} \stackrel{f(0)=0}{=} \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x} \stackrel{\text{def}}{=} f'(0) = f(1) = g(0)$, which implies that g is also right-continuous at $x_0 = 0$. It follows that g is continuous on $[0, 1]$. Note also that g is differentiable on $(0, 1)$ by AOD. As $g(0) = g(1) = f(1)$, then by Rolle's theorem there exists a point $0 < c < 1$ for which $g'(c) = 0$. We now differentiate for every $0 < x < 1$ using AOD,

$$g'(x) = \frac{f'(x) \cdot x - f(x)}{x^2}$$

Plugging the value $x = c$ we obtain

$$0 = g'(c) = \frac{f'(c) \cdot c - f(c)}{c^2}$$

which implies that $f'(c) \cdot c = f(c)$. Dividing by c yields $f'(c) = \frac{f(c)}{c}$.

3.2. (8 points) Find all values of $\alpha \geq 0$ for which the following series is convergent: $\sum_{n=1}^{\infty} \left(\frac{\alpha n + 1}{n + 1}\right)^n$.

Solution:

We use the root test,

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{\alpha n + 1}{n + 1}\right)^n} = \lim_{n \rightarrow \infty} \frac{\alpha n + 1}{n + 1} = \lim_{n \rightarrow \infty} \frac{\alpha + \frac{1}{n}}{1 + \frac{1}{n}} = \alpha$$

It follows that the series converges for $0 \leq \alpha < 1$ and diverges from the necessary criterion for $\alpha > 1$. If $\alpha = 1$ then we obtain the series $\sum_{n=1}^{\infty} 1^n = \sum_{n=1}^{\infty} 1 = \infty$. Thus, the series is convergent if and only if $0 \leq \alpha < 1$.

4. (25 points)

4.1. (18 points) Consider the function defined by $f(x) = \frac{1}{2} \left(x + \frac{2}{x}\right)$ for every $x > 0$, and consider the sequence $(a_n)_{n=1}^{\infty}$ defined by

$$\begin{cases} a_{n+1} = f(a_n) & \forall n \in \mathbb{N} \\ a_1 = 2 \end{cases}$$

4.1.1. (6 points) Compute the minimal value of f or prove that it doesn't exist.

Solution: Note that f is differentiable by AOD and for every $x > 0$ we have

$$f'(x) = \frac{1}{2} \left(1 - \frac{2}{x^2} \right) = \frac{1}{2} \frac{x^2 - 2}{x^2} = 0 \Leftrightarrow x = \sqrt{2}$$

As $f'(x) \geq 0$ for every $x \geq \sqrt{2}$ and $f'(x) \leq 0$ for every $0 < x \leq \sqrt{2}$ then f is decreasing on $(0, \sqrt{2}]$ and increasing on $[\sqrt{2}, \infty)$. Thus, $x = \sqrt{2}$ is a minimum of f , and the minimal value is $f(\sqrt{2}) = \sqrt{2}$.

4.1.2. (7 points) Prove that the sequence $(a_n)_{n=1}^{\infty}$ is well defined, and that it is convergent.

Solution:

As $a_1 = 2 > \sqrt{2}$ and for every $n \in \mathbb{N}$ we have $a_{n+1} = f(a_n) \geq \sqrt{2}$ then $a_n > 0$ for all $n \in \mathbb{N}$, and therefore the sequence $(a_n)_{n=1}^{\infty}$ is well defined and bounded from below by $\sqrt{2}$. We now prove by induction that the sequence $(a_n)_{n=1}^{\infty}$ is decreasing. First, we have $a_2 = 1.5 \leq 2 = a_1$. Let $n \in \mathbb{N}$ and suppose that $\sqrt{2} \leq a_{n+1} \leq a_n$. As f is increasing on $[\sqrt{2}, \infty)$ then

$$a_{n+2} = f(a_{n+1}) \leq f(a_n) = a_{n+1}$$

Thus, $(a_n)_{n=1}^{\infty}$ is bounded from below and decreasing, and therefore $\lim_{n \rightarrow \infty} a_n$ exists.

4.1.3. (5 points) Compute $\lim_{n \rightarrow \infty} a_n$.

Solution: As the sequence $(a_n)_{n=1}^{\infty}$ is convergent then there exists an $L \in \mathbb{R}$ such that $\lim_{n \rightarrow \infty} a_n = L$. As $a_n \geq \sqrt{2}$ for every $n \in \mathbb{N}$ then by the Monotonicity of limits we have $L \geq \sqrt{2}$. Taking the limits of both sides of the equation $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$ we obtain

$$L = \frac{1}{2} \left(L + \frac{2}{L} \right)$$

which implies that $L^2 = 2$. Thus, $L = \sqrt{2}$.

4.2. (8 points) Find a function $f : (-1, 1) \rightarrow \mathbb{R}$ for which

$$f^{(n)}(0) = (-1)^n \cdot (n+1)! \quad \forall 0 \leq n \in \mathbb{Z}$$

Note: you may leave your answer as a series.

Solution:

Solution: Consider the power series $\sum_{n=0}^{\infty} \frac{(-1)^n(n+1)!}{n!} x^n = \sum_{n=0}^{\infty} (-1)^n(n+1)x^n$. Note that the convergence radius is given by

$$R = \frac{1}{L} = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n+2} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{1 + \frac{2}{n}} = 1$$

Thus the series converges on the open interval $(-1, 1)$. Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n(n+1)!}{n!} x^n$ for every $x \in (-1, 1)$. Then f is well defined and therefore, by the uniqueness theorem, for every $n \geq 0$ we have

$$\frac{f^{(n)}(0)}{n!} = a_n \Rightarrow f^{(n)}(0) = n! \cdot a_n = (-1)^n(n+1)!$$

5. (25 points)

5.1. (8 points) Prove that $e^x \geq 1 + x + \frac{x^2}{2}$ for every $x \geq 0$.

Solution: Consider the function defined by $f(x) = e^x - 1 - x - \frac{x^2}{2}$ for every $x \geq 0$. Note that f is differentiable by AOD and that $f'(x) = e^x - 1 - x$ for every $x \geq 0$. As we have proved in class that $e^x \geq 1 + x, \forall x \geq 0$, then $f'(x) \geq 0$ and therefore f is increasing. Thus, for every $x \geq 0$ we have $f(x) \geq f(0) = 0 \Rightarrow e^x \geq 1 + x + \frac{x^2}{2}$.

An alternative solution: first, for $x = 0$ the inequality holds as an equality. Let $x > 0$. By Taylor's remainder formula, there exists a point $0 < c < x$ such that

$$R_2(x) = \frac{f^{(3)}(c)}{3!} x^3 = \frac{e^c}{6} x^3 > 0$$

Note that the Taylor polynomial centered at $x_0 = 0$ of order 2 is $T_2(x) = 1 + x + \frac{x^2}{2}$. Thus

$$e^x = T_2(x) + R_2(x) > T_2(x) = 1 + x + \frac{x^2}{2}$$

An alternative alternative solution: Let $x \geq 0$. Then

$$e^x - 1 - x - \frac{x^2}{2} = \sum_{n=0}^{\infty} \frac{x^n}{n!} - 1 - x - \frac{x^2}{2} = \sum_{n=3}^{\infty} \frac{x^n}{n!} \geq 0$$

where the last inequality follows from the fact that a nonnegative convergent series must converge to a nonnegative value.

5.2. (9 points) Prove that

$$\frac{1}{7\sqrt{2}} \leq \int_0^1 \frac{x^6}{\sqrt{1+x^2}} dx \leq \frac{1}{7}$$

Solution: First, as the function $\frac{x^6}{\sqrt{1+x^2}}$ is continuous on $[0, 1]$ by AOC then it is integrable, and therefore the integral $\int_0^1 \frac{x^6}{\sqrt{1+x^2}} dx$ exists. By the Monotonicity of the definite integral,

$$\int_0^1 \frac{x^6}{\sqrt{1+x^2}} dx \leq \int_0^1 \frac{x^6}{1} dx = \left[\frac{x^7}{7} \right]_0^1 = \frac{1}{7}$$

and

$$\int_0^1 \frac{x^6}{\sqrt{1+x^2}} dx \geq \int_0^1 \frac{x^6}{\sqrt{2}} dx = \frac{1}{\sqrt{2}} \left[\frac{x^7}{7} \right]_0^1 = \frac{1}{7\sqrt{2}}$$

5.3. (8 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive numbers and let $p > 1$. Suppose that $\lim_{n \rightarrow \infty} [\ln(a_n) + p \ln(n)]$ exists. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Solution: As the limit $\lim_{n \rightarrow \infty} [\ln(a_n) + p \ln(n)]$ exists, then there exists an $\alpha \in \mathbb{R}$ for which

$$\alpha = \lim_{n \rightarrow \infty} [\ln(a_n) + p \ln(n)] = \lim_{n \rightarrow \infty} [\ln(a_n) + \ln(n^p)] = \lim_{n \rightarrow \infty} \ln(a_n \cdot n^p) = \lim_{n \rightarrow \infty} \ln \frac{a_n}{\left[\frac{1}{n^p}\right]}$$

It follows that $\lim_{n \rightarrow \infty} \frac{a_n}{\left[\frac{1}{n^p}\right]} = \lim_{n \rightarrow \infty} e^{\ln\left(\frac{a_n}{\left[\frac{1}{n^p}\right]}\right)} = e^\alpha > 0$ and therefore the series $\sum_{n=1}^{\infty} a_n$ is convergent together with the p -Harmonic series (which converges as $p > 1$).